Multichannel Weak Signal Extraction Based on Multispectral LiDAR

Xiaxia Hou, Shalei Song, Yanhong Ran, Yuxuan Chen, Zhenwei Chen, Binhui Wang, and Long Guo

***Abstract*— The information extracted from waveform data of full-waveform light detection and ranging (LiDAR) has been widely used in applications such as 3-D urban modeling, target recognition, and classification. However, the presence of weak signals is inevitable in LiDAR systems. To enhance its effective detection capability and extraction accuracy, we propose a multispectral LiDAR (MSL) weak signals extraction (MSL-WSE) method. The measurement data from our MSL system were used to evaluate the performance of the proposed method. The correlation coefficient (R^2), root mean square error (RMSE), and effective extraction rate show that the MSL-WSE method accurately detected and extracted the waveform parameters of weak echo signals, providing a more realistic and fine-grained true color 3-D point cloud.**

***Index Terms*— Adaptive threshold, correlation analysis, multispectral light detection and ranging (MSL), wave information detection and extraction, weak signal.**

I. Introduction

LIGHT detection and ranging (LiDAR) is widely used in target imaging [1], terrain measurement [2], 3-D reconstruction of buildings [3], [4], and other fields due to its unique advantages such as high accuracy, large-scale detection, and high-density point cloud acquisition. LiDAR uses emitted laser beams to interact with targets and invert their distance, shape, surface characteristics, and other features from the resulting echo signals. However, due to limitations of the system and the environment, the propagation of LiDAR echo signals in practical applications is proportional to the square of distance and thus attenuates with distance. Weak echo signals often appear in the far-field, leading to partial loss of target information and consequently affecting the quality of point cloud visualization [5], [6]. With the development of LiDAR technology and the demand for 3-D true color visualization, LiDAR is moving toward single wavelength to multispectral or even hyperspectral, to achieve integrated acquisition of spatial and spectral information [7], [8], [9], [10].

This new kind of multispectral LiDAR (MSL) uses a supercontinuum laser as a light source [11], [12], [13], [14]. The energy distribution of this wide-spectrum laser is uneven, and the energy of each band is generally only one-tenth of the traditional single-wavelength LiDAR. Furthermore, supercontinuum laser is typically pumped in the infrared band, resulting in relatively weak transmitted energy in the short-wave bands which are far away from the pump light [15]. While for, ground objects, especially vegetation, usually have lower reflectance in the visible band than in the near infrared. All these factors will cause the echo energy of MSL to be more prone to weak signals, especially in the visible band. This poses a crucial challenge in extracting echo signals from weak channels of this new kind of MSL system.

LiDAR signals exhibit characteristics of amplitude continuity and random phase over time, making them typical nonlinear and nonstationary signals due to the presence of noise [16]. Various denoising methods have been applied to process weak echo signals in single-wavelength LiDAR, including variational mode decomposition [16], low-pass filtering [17], wavelet transform (WT) [18], [19], Fourier transform [20], [21], and power spectrum estimation [22]. Research has shown that these methods have certain capabilities for denoising single-channel LiDAR. In MSL, the noise characteristics differ among different receiving channels, and the amplitudes of transmitted and echoed pulses within the same channel may vary [23], [24], [25], necessitating separate analysis. Building upon signal quality improvement, traditional signal extraction methods include thresholding [26], [27], peak detection [28], fixed timing algorithms [29], and deconvolution [30], [31]. Thresholding determines whether a signal exceeds a certain amplitude based on a predefined threshold, but its detection capability for weak signals is limited [28]. Peak detection methods can identify the amplitude and central position of normal signal pulses. However, in weak signal scenarios, there can be a certain level of error introduced due to the presence of noise interference. Fixed timing algorithms determine the presence of signals based on fixed time intervals, but they are not sensitive to the temporal characteristics of weak signals. Deconvolution aims to restore the original shape of a signal through inverse convolution operations, but its recovery

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TABLE I
BAND NUMBER AND CORRESPONDING CENTER WAVELENGTH

Channel number	CH1	CH2	CH3	CH4
Center wavelength/nm	475	539	640	820

performance may be compromised by noise in weak signal situations [28].

In this article, we propose an MSL weak signals extraction (MSL-WSE) method by considering the multichannel characteristics of MSL. The filtering option of each weak echo is adaptively chosen to avoid the limitations of a single filter parameter. Then, the signals of each channel are divided into transmit and echo windows, and an adaptive threshold coefficient to compensate for the defect of weak signal leakage caused by a unified threshold. On this basis, the timing correction of multichannel is carried out by analyzing the correlation between each channel, to determine the confidence interval and confidence channel, which can improve the shortcomings of the existing single-channel weak echo signal recognition. The method ultimately improves the recognition and extraction capability of weak echo signals, thereby enhancing the quality of the 3-D point cloud and improving its visualization effects. The objectives of this study are as follows.

1) Proposes an MSL-WSE method for detecting and extracting weak echo signals in MSL data.
2) A time alignment strategy is proposed, which utilizes correlation analysis between channels and the peak position of the transmitted pulses to eliminate waveform time offsets between channels.
3) A dynamic optimization cumulation strategy based on multichannel weak echo correlation is adopted to improve the effective extraction rate for each channel.

II. MATERIALS

In recent years, we have been dedicated to researching MSL. Building upon the foundation of the previous three wavelength LiDAR systems [23], [25], [32], red (R), green (G), and blue (B) bands combination were selected to realize a true color 3-D visualization based on the energy distribution of the supercontinuum laser source, detector response, and chromaticity principles. Besides, a near-infrared (NIR) band has been added which provides the capability of vegetation recognition through the setup of normalized difference vegetation index (NDVI) [10], [33]. The center wavelengths of our MSL are shown in Table I. The system's light source uses the new SuperK COMPACT supercontinuum (SCS) white light source, which has a wide output spectrum covering the entire 450–2400 nm region, with a total output power of more than 110 mW and an output power stability of ±1.5% (1 h). The laser operates at a repetition rate of 22 kHz with an average power of 100 mW and a pulsewidth of 2 ns.

Due to the wide spectrum range coverage, the pulse energy emitted by a supercontinuum laser would exhibit more obvious fluctuation than the single wavelength laser source, which will increase the uncertainty of echo intensity detection [34]. Therefore, the transmitting and receiving waveforms of each channel are recorded simultaneously to eliminate the fluctuations and influence of each pulse. The waveforms of the four channels are digitized by a 12-bit analog-to-digital converter with a sampling rate of 3.2 GHz. Fig. 1(a) shows the MSL system from which the R, G, B, and NIR intensity of real scenes could be achieved by 3-D scanning. To be clear, the MSL system also exhibits weak interchannel correlation according to wavelength selection theory [23], which helps to avoid the potential risk of weak signals occurring in all channels as much as possible.

Fig. 1. MSL system and experimental objects. (a) MSL system. (b) Color checker. (c) Mural.

Two experiments were designed to verify the effectiveness of the weak signal extraction method proposed in this article [see Fig. 1(b) and (c)]. First, using a standard Commission Internationale de l'Eclairage (CIE) color checker to evaluate the feasibility and accuracy of the proposed method, and then a real scenario detection model was conducted to determine the applicability of the method. For Fig. 1(b), the nonlinear RGB values of each color block are known and widely used for color calibration of digital cameras, scanners, and other devices under the CIE standard light source D65 [35]. At 26 m, the laser spot hits the color checker (29 × 20.3 cm) vertically and shifts in sequence according to the sequence of color blocks. The 60 000 pulses are repeatedly collected for each color block, with the laser-transmitted energy set to 5% to capture weak echo signals.

For Fig. 1(c), the mural (99.5 × 74 cm) with a dark overall tone and coated with anti-corrosion material was chosen as the second experimental scene to further verify the weak echo signal extraction capability. These two experiments reproduce the scene of weak echo signals which is easy to appear when MSL detects the targets with low-reflection materials and complex colors. These two representative scenarios are tried to verify the effectiveness of the multichannel weak signal extraction method.

III. METHODS

The MSL system simultaneously records the backscattered waveforms of the RGB channels as well as the NIR channel, which allows for the extraction of color and spatial information for each echo signal without interpolation, enabling direct visualization of targets as true-color 3-D point clouds. Therefore, ensuring the accurate retrieval and extraction of weak echo signals in each channel is crucial. Temporal consistency exhibited by the echoes from the same target in all channels lays the foundation for interchannel correlation detection. To recognize and extract weak echo signals in MSL, we propose an MSL-WSE method, as illustrated in Fig. 2. It consists of three steps: multichannel weak signal preprocessing (MC-WSP), multichannel weak signal detection (MC-WSD), and multichannel weak signal parameter extraction (MC-WSPE).

A. Multichannel Weak Signal Preprocessing (MC-WSP)

Considering the characteristics of noise, it is common practice to use waveform data that does not contain valid echo signals to estimate the noise level. The noise threshold is usually set to three times the standard deviation of the

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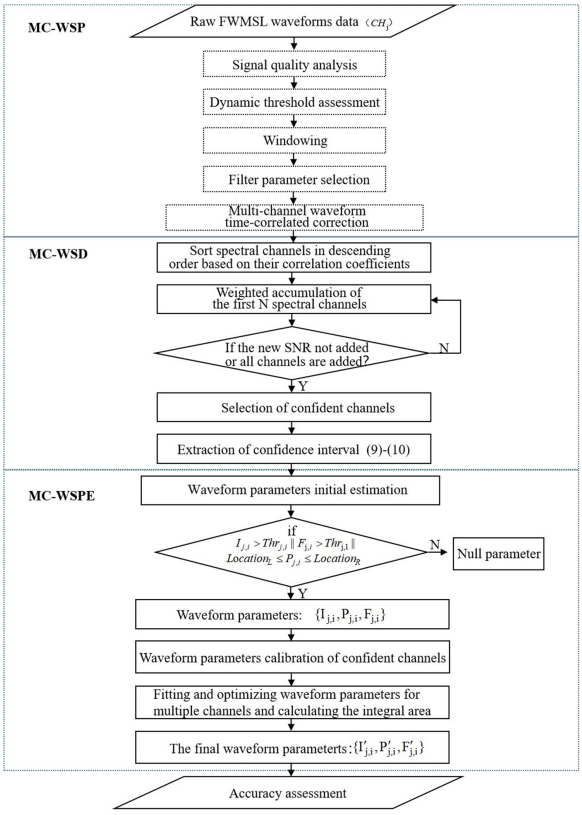

Fig. 2. MSL weak echo processing flowchart, including MC-WSP, MC-WSD, and MC-WSPE.

background noise [36], [37]. However, this fixed threshold would cause missed detections due to the overestimation of the weak signals. In this study, a dynamic threshold $\mathrm{Thr}_{j,i}$ was made to calculate the thresholds for the transmitted waveform and target echoes separately, aiming to detect weak echo signals as much as possible as in the following equation:

$$\mathrm{Thr}_{j,i} = kS_{j,i} + M_{j,i}. \tag{1}$$

Assuming there are j receiving channels, each channel contains i waveforms, where $i = 1$ represents the transmitted pulse, and $i = 2, \ldots, n$ represents the echo signals. Meanwhile, all backscatter waveforms recorded by each channel are divided into a transmit window and an echo window to reduce data redundancy and extract confidence intervals. $S_{j,i}$ represents the mean square deviation of noise for the i waveform window in the j channel. $M_{j,i}$ represents the average value of noise. k represents the adaptive threshold coefficient. Therefore, the echo pulse that has a peak value is less than or equal to $\mathrm{Thr}_{j,i}$ can be defined as a weak signal.

In addition, an optimal filter needs to be selected to improve the signal quality of weak echoes based on MSL multichannel. In MSL multichannel backscattered waveforms, noise is often considered as Gaussian white noise, making Gaussian filters well-suited for noise matching [24]. In addition, the noise amplitude in weak echoes is generally higher and accompanied by higher frequency components. Savitzky-Golay (S-G) filters can effectively smooth the signal while preserving its shape and details [38]. The WT denoising methods can also analyze the signals at multiple scales to balance the denoising effect and signal preservation [6], [39]. The multichannel signals of MSL exhibit nonlinearity and nonstationarity, and its data volume is also several times that of traditional single-wavelength LiDAR. Thus, selecting a filter that is suitable for the characteristics of multichannel signals is a crucial prerequisite for MSL-WSE. By comparing the signal-to-noise ratio (SNR), root mean square error (RMSE), and correlation coefficient (P), the performance of the aforementioned filters can be evaluated, and the most effective filtering method for MSL multichannel weak backscattered waveforms can be selected

$$\mathrm{RMSE} = \left(\frac{1}{N} \sum_{n=1}^{N} \left| y_{j,i}(t) - y'_{j,i}(t) \right|^2 \right)^{1/2} \tag{2}$$

$$p = \frac{\sum_{n=1}^{2N-1} y_{j,i}(t) y'_{j,i}(t)}{\left(\sum_{n=1}^{2N-1} y_{j,i}(t)^2 \cdot \sum_{n=1}^{2N-1} y'_{j,i}(t)^2 \right)^{1/2}} \tag{3}$$

where $y_{j,i}(t)$ represents the raw signal and $y'_{j,i}(t)$ represents the filtered signal. The SNR is positively correlated with the quality of weak backscattered signals.

For multichannel signals, it will first face the problem of the timing difference between channels which is reflected in the optical path difference of each channel, and the different response times of visible and NIR detections. The timing difference makes multichannel signals that cannot be accumulated directly to extract weak signals. It is necessary to establish a timing synchronization procedure between multichannel to eliminate the timing difference. The channel with the largest SNR value is taken as the reference channel, while the other receiving channels are cross correlated [see (4) and (5)] to it. As shown in (6), the Lag_j represents the reference time between the reference channel and other channels. Then, the reference time is used to unify the channels in the time domain, so that the location distribution of each channel $y_j(t)$ can be integrated into the subsequent signal accumulation

$$\rho_j(\tau) = \frac{y_j(t) \cdot y_j(t)}{2} \cos(\omega_0 \tau + \varphi - \theta) \tag{4}$$

where $\rho_j(\tau)$ represents the cross correlation function. $\omega_0 \tau$ represents frequency, and $\varphi - \theta$ is the phase difference. The cross correlation coefficients are represented as R

$$R = \begin{pmatrix} 1 & \rho_{1,2}(\tau) & \rho_{1,3}(\tau) & \rho_{1,4}(\tau) \\ \rho_{2,1}(\tau) & 1 & \rho_{2,3}(\tau) & \rho_{2,4}(\tau) \\ \rho_{3,1}(\tau) & \rho_{3,2}(\tau) & 1 & \rho_{3,4}(\tau) \\ \rho_{4,1}(\tau) & \rho_{4,2}(\tau) & \rho_{4,3}(\tau) & 1 \end{pmatrix} \tag{5}$$

$$y'_j = y_j(t) \pm \mathrm{Lag}_j \tag{6}$$

where y'_j represents the synchronized waveform signal of channel j. Besides, R can also be used for the calculation of the confidence channel.

B. Multichannel Weak Signal Detection (MC-WSD)

As we mentioned above, the probability of the weak signal occurrence in different channels of MSL is much higher than that of traditional single wavelength LiDAR. Especially, the contamination of echo noise which introduces certain errors make it hard to extract the real waveform parameters directly. The signal is usually initially enhanced by multichannel accumulation, the simple accumulation of multichannel is not the optimal solution, especially the extremely low SNR channels will even reduce the overall signal quality. In this article, a dynamic accumulation strategy based on the determination

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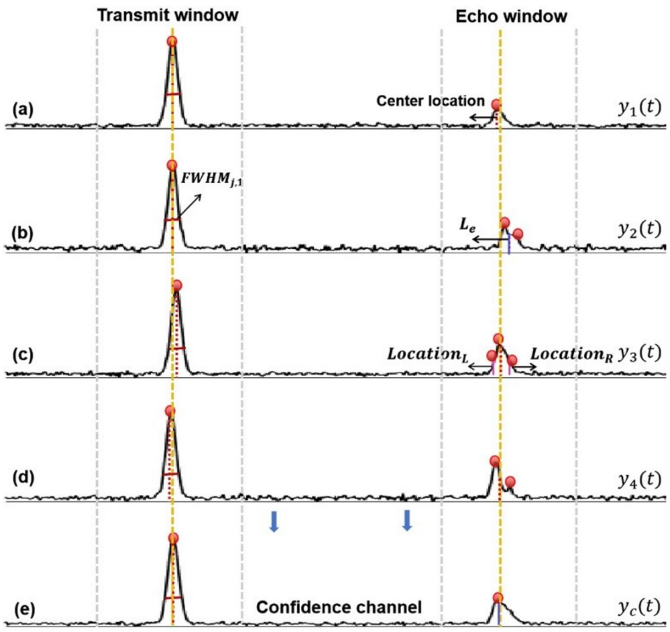

Fig. 3. Schematic of waveform and target echo confidence interval. (a)–(e) Represent the B, G, R, NIR and confidence channels respectively.

of confidence channel by using cross correlation coefficients is proposed to achieve an optimized signal accumulation effect. The selection iteration of accumulative channels is dependent on the R-value from high to low and stopped until the final SNR reaches the maximum. A new accumulation result $y_c(t)$ can be obtained by the following equation, which is considered as a confidence channel [see Fig. 3(e)]:

$$y_c(t) = \sum_{j=1}^{N} R * y_j(t). \tag{7}$$

The intensity and pulsewidth of the accumulated confidence channel are enhanced and broadened compared to a single receiving channel, but its center position remains determined. Therefore, in this article, only the center position of the confidence channel is taken as an input parameter to locate and correct the position of weak echo signals in the receiving channels. Intensity and pulsewidth are not considered in our analysis.

Even so, the confidence channel is still not a sufficient condition to direct and recognize the weak signals in each channel. The factors that generate the weak signals of MSL are very complicated. In addition to the target reflectance and the uneven energy distribution of the supercontinuum laser source, the scattering cross section, and the response efficiency of different types of detectors, would also result in the echo waveform shapes do not strictly follow Gaussian symmetry and even having a tail. These factors will lead to inaccurate positioning of the center position of the echo waveform, thus increasing the extraction error of weak echo parameters. As shown in Fig. 3, the yellow dashed line represents the initial position of weak echo indicated by the peak point position of the confidence channel, and the center position [see Fig. 3(a)] of each channel echo exhibits obvious variation.

Under complicated circumstances, it is necessary to confirm the weak signal range which we call the confidence interval before the identification of the center position. According to the asymmetry of the rising time and falling time of the weak echo signal, the confidence interval can be defined as $[\text{Location}_L, \text{Location}_R]$ [see (8) and (9)]. L_e represents the delay of the echo signal relative to the transmitted pulse on the time axis, which is obtained by using the autocorrelation function to find similar characteristics of the echo signal and the transmitted pulse [see Fig. 3(b)]. The echo signal is the result of the interaction between the transmitted pulse and the target along the optical path, and the full width at half maximum (FWHM) of the transmitted pulse remains relatively constant. Therefore, this article combines the FWHM of the transmitted pulse [see Fig. 3(b)] to calculate the confidence interval [see Fig. 3(c)] and then discriminates the echo waveform parameters extracted from each channel

$$\text{Location}_L = |L_e| - \frac{\text{FWHM}_{j,1}}{2} \tag{8}$$

$$\text{Location}_R = |L_e| + \frac{\text{FWHM}_{j,1}}{2} \tag{9}$$

$$R_j(L_e) = \frac{\sum (y_j(t) * y_j(t + L_e))}{N}. \tag{10}$$

$R_j(\text{Lag}_e)$ is the autocorrelation coefficient. $y_j(t)$ and $y_j(t + L_e)$, respectively, represent the sequence of waveform sampling points in transmit and echo windows, where t is the index of the sequence and N is the length of the sequence.

C. Multichannel Weak Signal Parameter Extraction (MC-WSPE)

After the procedure of preprocessing and MC-WSD, weak signals can be initially located and identified. However, it is not enough to ensure that the waveform parameters of weak signals can be accurately and effectively extracted, due to their low SNR and incomplete symmetrical shape. As shown in Fig. 3, there are significant differences in waveform shapes within the echo window. If the center position of the echo is determined solely by combining the threshold value and extreme points [26], single echoes with tails or multiple peaks may be misjudged as multiple echoes. This may potentially lead to local optimization in Levenberg-Marquardt (LM) and result in increased ranging and target recognition errors. Other discriminant conditions need to be introduced to screen the extracted maximum value and its corresponding position. Based on the confidence channel, the number of echo signals can be predetermined, and L_e can accurately locate the center of the weak echo in each channel. Moreover, the false echo waveform parameters that do not fit the range could be eliminated through confidence interval and threshold $\text{Thr}_{j,i}$ discrimination, as shown in the following equations:

$$I_{j,i} = \begin{cases} I_{j,i}, & I_{j,e} \geq \text{Thr}_{j,i} \\ 0, & I_{j,e} < \text{Thr}_{j,i} \end{cases} \tag{11}$$

$$P_{j,i} = \begin{cases} P_{j,i}, & \text{Location}_L \leq P_{j,e} \leq \text{Location}_R \\ 0, & P_{j,e} < \text{Location}_L || P_{j,e} > \text{Location}_R \end{cases} \tag{12}$$

$$F_{j,i} = \begin{cases} F_{j,i}, & F_{j,e} \geq \text{Thr}_{j,i} \\ 0, & F_{j,e} \leq \text{Thr}_{j,i} \end{cases} \tag{13}$$

where $I_{j,e} P_{j,e}, F_{j,e}$ represent the intensity, central position, and FWHM of the initial estimated echo signal, respectively. $I_{j,i} P_{j,i}, F_{j,i}$ are echo waveform parameters that satisfy the discriminant conditions. Then, the initialized waveform parameters are optimized and fit using the multichannel Gaussian

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mixture model [23] and the LM algorithm [40], until the fitting accuracy R_j^2 reaches its maximum, as shown in the following equation:

$$y'_{j,i}(t) = \sum_{i=1}^{N} I_{j,i} \exp\left\{ -\frac{(t - P_{j,i})^2}{F_{j,i}^2/(4 \ln 2)} \right\} + \text{noise}_{j,i}. \quad (14)$$

The amplitude errors caused by energy fluctuation can be reduced by calculating the integrated area (A) of weak echo. The intensity of the waveform is represented by its integrated area $I'_{j,i}$, where the optimized parameters $P'_{j,i}$ and $F'_{j,i}$ represent the center position and FWHM. Finally, the echo waveform parameters ($\{I'_{j,i}, P'_{j,i}, F'_{j,i}\}$) from all channels can be extracted.

The intensity value of the extracted echo is directly proportional to the linear RGB value [41]. When using MSL, the waveform decomposition is performed by the method proposed in this article, and the intensity ratio of the echo signal to the transmitted pulse within each receiver channel is calculated to obtain linear RGB values. Considering the complexity of the actual measurement environment, we further compare with targets with known reflectivity in the test area to obtain more accurate RGB values. A gamma correction is applied to convert it into a nonlinear value to avoid distortion. By combining the scanning angle, distance, and RGB value from each channel, the true color 3-D point cloud of the target can be obtained [42].

D. Accuracy Assessment

Correlation coefficient (R^2) and RMSE are used as linear evaluation metrics to assess the accuracy of the extracted RGB values from the color checker. Higher R^2 and lower RMSE indicate a closer approximation to the true values. While, due to the nonuniform perceptual nature of the RGB color space, these two metrics cannot assess the human eye's perception of color. Thus, the color difference ΔE (15) is introduced, which is typically calculated in the uniformly perceptually uniform LAB color space [43]

$$\Delta E = \sqrt{(L_1^2 - L_2^2) + (A_1^2 - A_2^2) + (B_1^2 - B_2^2)} \quad (15)$$

where L_1 and L_2 are the lightness indices, and A_1, A_2, B_1, and B_2 are the chromaticity indices. As the ΔE value decreases, it shows that the full-color visualization of the point cloud is presented with greater realism and accuracy.

In response to the situation fact where it is hard to obtain the true RGB true value of the mural as a reference for verification is unknown, the effective extraction rate (the ratio of effective extraction points to the true sampling points) was introduced to characterize the extraction ability of weak echoes in our study.

IV. Results and Discussion

A. Improvement of Multichannel Weak Echo Data Quality and Temporal Calibration

In Section II, experiments were designed by using a color checker and mural to evaluate the performance and accuracy of weak echo waveform parameters extracted by the WSL-WSE method, and their impact on the point cloud quality. As the color checker comprises 24 color blocks, the reflectance of the first color block is relatively low, and there is a weak echo signal, making it more representative. Therefore, it is chosen to demonstrate the processing performance of the algorithm proposed in this article. In addition, the scanning data of all color blocks from the color checker are used to test the robustness of the method proposed in this article. The visual results and quantitative comparison outcomes are provided in Section IV-B.

TABLE II
Wavelet Denoising Filter Effect of Different Channels

	Wavelength			
	475nm	539nm	640nm	820nm
Raw SNR (dB)	0.0058	0.8773	11.3517	10.9524
Fitted SNR (dB)	4.3389	6.0694	16.5965	12.7205
RMSE	0.5584	0.6541	0.5744	0.1818
P	0.9114	0.8407	0.9723	0.9984
level	3	3	3	3
function	sym6	sym3	db6	sym4

As shown in Table II, there were significant differences in the SNR of each receiving channel, especially in the shortwave channels of 475 and 539 nm, which were less than 1 dB. Consequently, it is necessary to select appropriate filters and filtering parameters for each channel based on their different noise level.

Adaptive selection of filters and their parameters is performed according to Section III-A. The results indicated that when $r = 5$, Sigma $= 3$ [see Fig. 4(a)–(c)], order $= 2, 3$, Framelen $= 13$ [see Fig. 4(d)–(f)], and level $= 3$, function $=$ sym, $N = 6$ [see Fig. 4(g)–(i)], Gaussian, S-G, and WT filters achieved well performance, while the WT filter yielding the highest SNR value. Thus, WT should be the best filter to remove high-frequency noise.

Table II summarizes the filtering parameters used in all channels, and the filtering results are shown in Fig. 5. The first and third rows display the raw and filtered waveforms of the four channels. The second and fourth rows showed their corresponding short-time Fourier transform (STFT) spectra. From the comparison of the original waveform and the filtered waveform, it could be seen that the MSL-WSE method effectively suppressed high-frequency noise by adaptively selecting filtering parameters. The filtered waveform exhibited smoother characteristics while preserving the details and structure of the signal. This shows that the MSL-WSE method effectively improved signal quality when dealing with weak signals.

The timing deviation between channels will affect the accuracy of weak echo recognition. Therefore, timing synchronization should be performed in all channels. In Fig. 6(a), we can find that there was a time deviation of up to 25 sampling points between the receiving channels before the timing synchronization, which limited the use of multichannel joint detection of weak echoes. However, after synchronous calibration, the transmitted pulses and target echoes of all channels were evenly distributed at t'_1 and t'_2, and the sampling point deviation was reduced to 0 samples [see Fig. 6(b)], indicating that the timing difference was effectively eliminated. This improvement in timing synchronization helps to detect the location of weak signals from the same target on different channels.

B. Multichannel Weak Echo Parameter Extraction and Validation

After the time synchronization calibration between channels, we compared the MSL-WSE method with the

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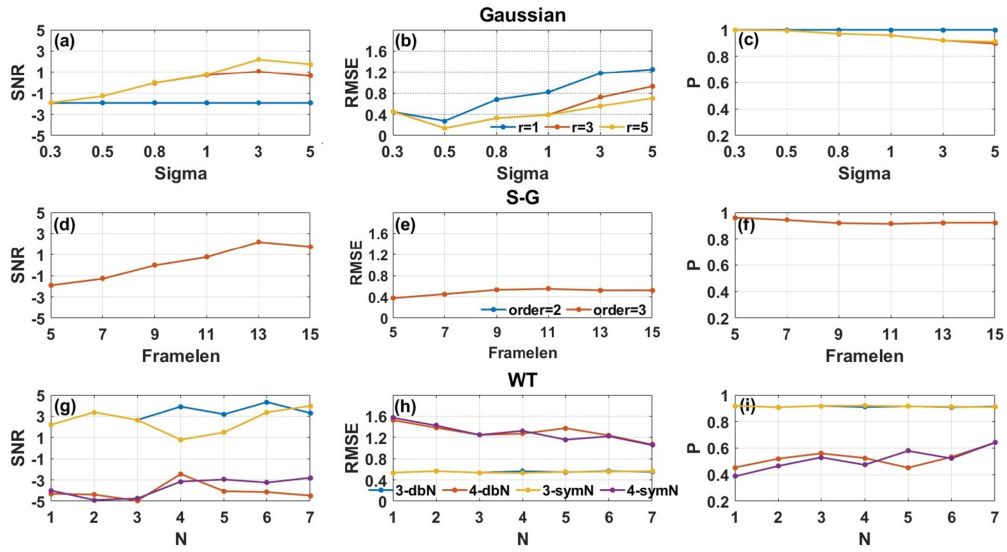

Fig. 4. Filter adaptive selection results. (a)–(c) Gaussian filtering, where Sigma and r represent the standard deviation and filter radius, respectively. (d)–(f) S-G filtering, where Framelen and order are the filter radius and order of the fitting polynomial, respectively. (g)–(i) WT, where N is the wavelet function vanishing moments, 3 and 4 are the levels of decomposition, respectively, db$N$ and sym$N$ are the wavelet functions.

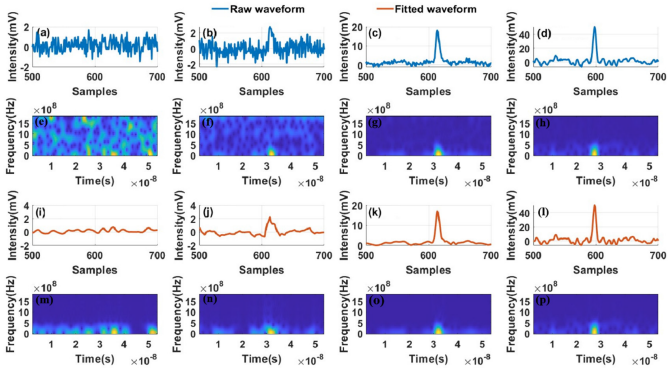

Fig. 5. Multichannel filter result. (a)–(d) and (i)–(l) Raw and filtered target echoes, respectively. (e)–(h) and (m)–(p) Raw and filtered STFT spectra of target echoes, respectively. (a) and (i) 475 nm. (b) and (j) 539 nm. (c) and (k) 640 nm. (d) and (l) 820 nm.

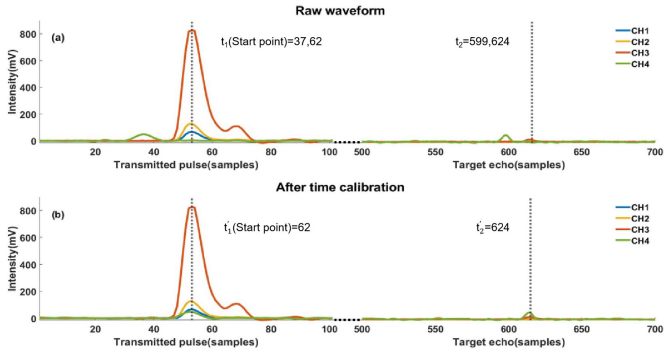

Fig. 6. Waveform display results before and after multichannel time synchronization calibration. (a) Raw waveforms of transmitted and target echo at each channel, while (b) results after multichannel time synchronization calibration.

traditional single-channel waveform decomposition (SCWD) method [26], [30] to evaluate its advantages in detecting and extracting weak echo signals, as well as their color restoration capabilities and visualization effects.

Figs. 7 and 8 displayed the results of the SCWD method and the MSL-WSE method in terms of echo signal detection and waveform fitting within each channel, respectively. The monochrome point clouds were obtained by inverting the corresponding waveform parameters from the respective receiving channels. In the high SNR channel with Gaussian symmetric waveform shape [see Fig. 7(c)], the waveform parameters detected by both SCWD and MSL-WSE methods were similar, and the subsequent waveform fitting results showed comparable accuracy between the two methods. However, in the channels with lower SNR, the SCWD method employed the fixed threshold that was set relatively high and disregarded the correlation between the receiving channels.

In Fig. 7(b), although the SCWD method detects two peak points above the threshold, their corresponding FWHMs are four and two samples, which are smaller than the emitted pulses (eight samples), and do not satisfy the filtering condition, resulting in the failure of waveform parameter extraction. Therefore, even if multiple peak points were identified, weak echo signals were still missed, leading to the A-value of 0 and negative fitting accuracy, as shown in Fig. 8(a) and (b). These unrecognized signals result in an RGB value of 0, making the point cloud appear darker in color. In contrast, the MSL-WSE method adaptively set thresholds based on the SNR in different channels and considered the consistent spatial distribution of the same target across multiple receiving channels. It utilized the high-amplitude signal localization from the confidence channel to locate weak signals through other channels. Therefore, the MSL-WSE method can accurately detect the weak echo signal and obtain the corresponding integral area by waveform fitting with high fitting accuracy (>0.9), which further advanced its accuracy in weak signal extraction, and finally made the point cloud show a bright blue–green color, as shown in Fig. 7(n)–(o).

In addition, due to the ringing effect of the detector, there may be trailing noise near the true echo signal with a similar FWHM to it. The waveform shape did not strictly adhere to Gaussian symmetry. The SCWD method only relied on the threshold and FWHM, could not effectively differentiate this

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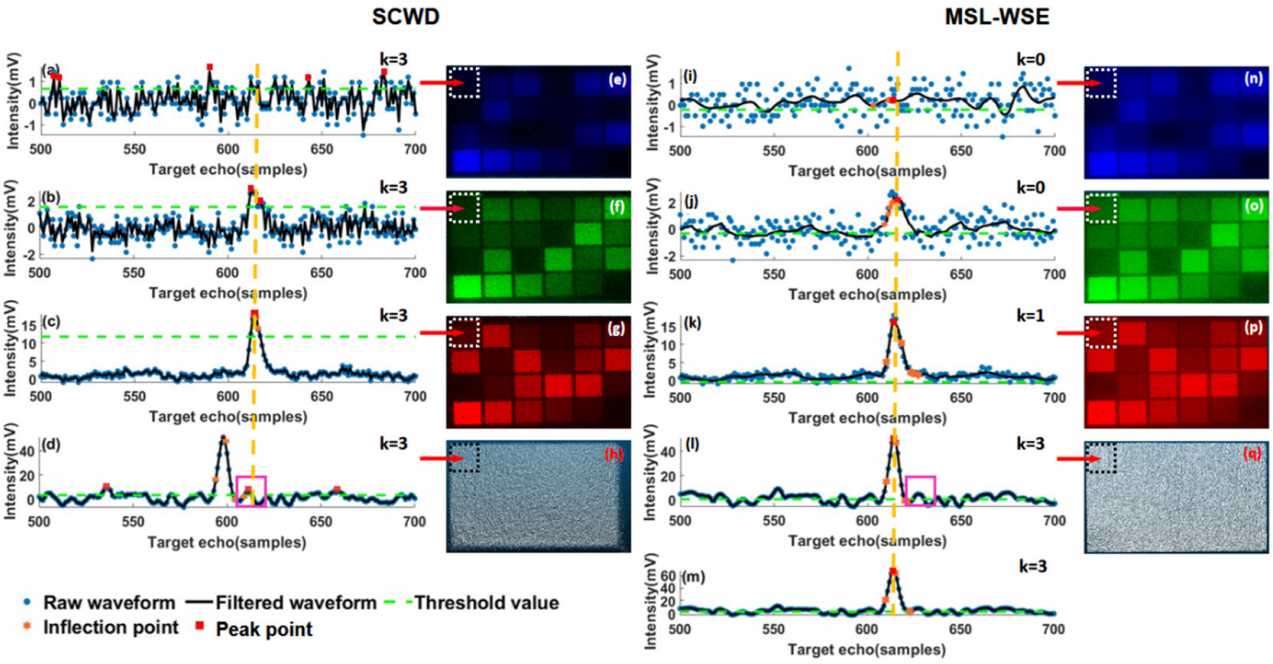

Fig. 7. Waveform parameter information obtained by SCWD and MSL-WSE methods separately, and the corresponding monochrome point cloud. Where the waveform of each receiving channel corresponds to an echo point in the white box, and k represents the threshold coefficient. In the 475 and 539 nm channels, the range of target echo signal amplitude is between 0 and 2.4 mV, in the 640 and 820 nm channels, it is between 0 and 50 mV. Distinguishingly, the 820 nm channel that is invisible to the human eye is represented by a grayscale point cloud. (a) and (i) 475 nm. (b) and (j) 539 nm. (c) and (k) 640 nm. (d) and (l) 820 nm. (m) Confidence channel. (e)–(h) and (n)–(q) Respectively represent the single-channel point clouds corresponding to the 475nm-820nm channels obtained using the SCWD method and the MSL-WSE method.

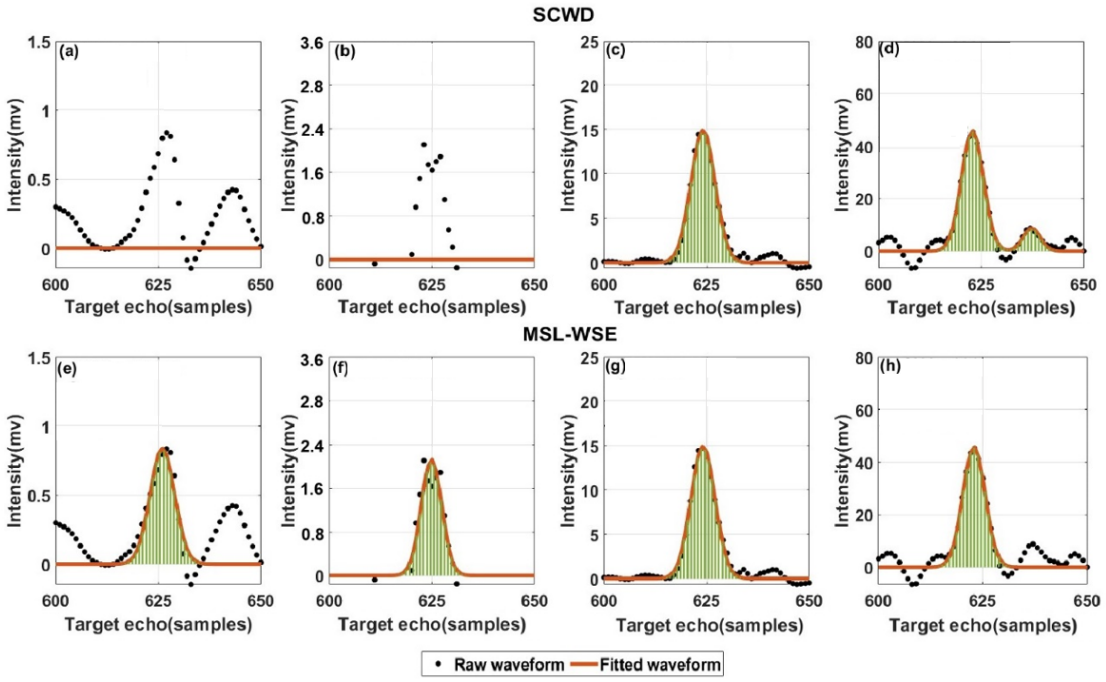

Fig. 8. Results of optimized waveform fitting obtained by performing SCWD and MSL-WSE at channels of 475, 539, 640, and 820 nm. Each subgraph consists of the raw waveform and the fit waveform, where R_j^2 represents the fitting accuracy and A represents the integrated area (green shadow) of the fit waveform. (a) CH1: 475 nm. $R_1^2 = -0.9004$. $A = 0$. (b) CH2: 539 nm. $R_2^2 = -0.1526$. $A = 0$. (c) CH3: 640 nm. $R_3^2 = 0.9830$. $A = 109.9172$. (d) CH4: 820 nm. $R_4^2 = 0.9498$. $A_1 = 306.3833$. $A_2 = 46.0805$. (e) CH1: 475 nm. $R_1^2 = 0.9058$. $A = 11.9988$. (f) CH2: 539 nm. $R_2^2 = 0.9527$. $A = 8.2955$. (g) CH3: 640 nm. $R_3^2 = 0.9859$. $A = 109.9172$. (h) CH4: 820 nm. $R_4^2 = 0.9134$. $A = 374.3200$.

trailing noise, and falsely detected it as a signal [red box in Fig. 7(d)], resulting in two sets of A-values in waveform fitting, as shown in Fig. 8(d). To address this issue, the MSL-WSE method obtains confidence channels based on the correlation between the received channels [see (4)–(6)], initially obtaining only one echo signal from the target under test [see Fig. 7(m)]. Subsequently, the confidence interval for the true echo signal is determined to be [619, 627] according

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TABLE III
WAVEFORM PARAMETERS OF TARGET ECHO WERE EXTRACTED BY SCWD AND MSL-WSE IN EACH CHANNEL

methods	N	475nm			539nm			640nm			820nm		
		$I_{1,i}$	$P_{1,i}$	F_{1i}	$I_{2,i}$	$P_{2,i}$	$F_{2,i}$	$I_{3,i}$	$P_{3,i}$	$F_{3,i}$	$I_{4,i}$	$P_{4,i}$	$F_{4,i}$
SCWD	N_1	0	0	0	0	0	0	109.92	624.02	6.50	306.38	623.80	6.26
	N_2	-	-	-	-	-	-	-	-	-	46.08	639.28	5.91
MSL-WSE	N_1	12.00	625.98	7.03	8.30	624.83	5.86	109.92	624.10	6.92	374.32	623.85	6.27
	N_2	-	-	-	-	-	-	-	-	-	-	-	-

N: Waveform identifications, $I_{j,i}$: Intensity (mV), $P_{j,i}$: Center position (samples), $F_{j,i}$: FWHM (samples).

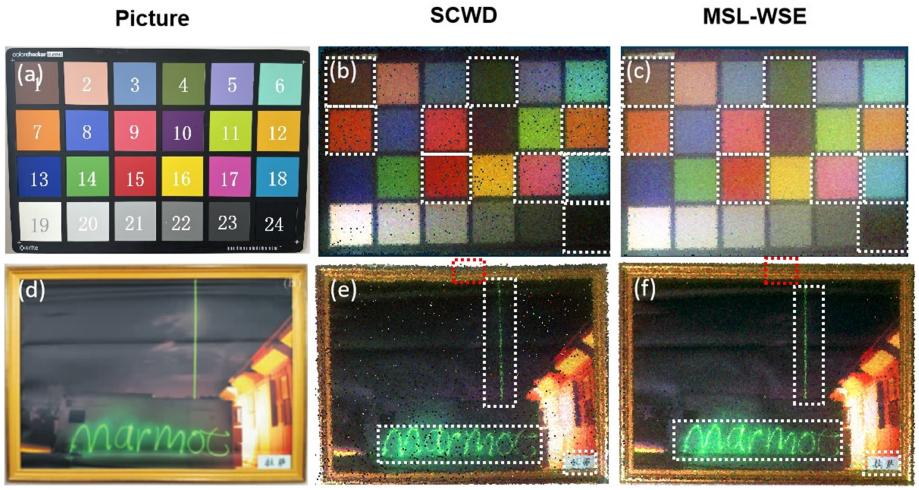

Fig. 9. Point cloud results of the color checker and mural obtained by the SCWD and MSL-WSE methods. (a) and (d) Pictures of the color checker and mural, respectively. (b), (c), (e), and (f) Respectively represent the color point clouds of the color checker and mural obtained using the SCWD and MSL-WSE methods.

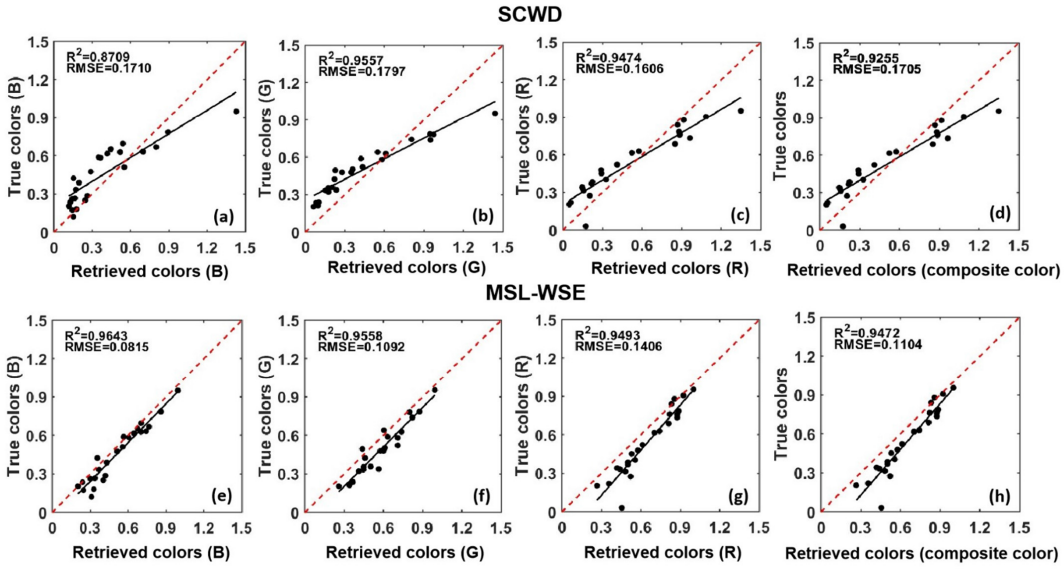

Fig. 10. Linear assessment results between retrieved colors and true colors obtained by the SCWD and MSL-WSE methods. (Left to right) B (475 nm), G (539 nm), R (640 nm), and composite color. (a)–(d) Respectively represent the correlation between the color values in the B, G, R, and composite color channels of the color checker obtained using the SCWD method and their true values. (e)–(h) Respectively represent the correlation between the color values in the B, G, R, and composite color channels of the color checker obtained using the MSL-WSE method and their true values.

to (8)–(10). Within this interval, we use the inflection point method to accurately detect a unique peak point with the waveform parameters {374.32, 623.85, 6.27}, which satisfies the discrimination conditions in (11)–(13) and is, therefore, considered as the waveform parameters of the target echo. The peak coordinate of the second pulse in the echo window is [639.28, 46.08], which falls outside the confidence interval. Therefore, it is regarded as trailing noise rather than a true echo signal and is not included in the subsequent waveform fitting process [see Figs. 7(q) and 8(h)].

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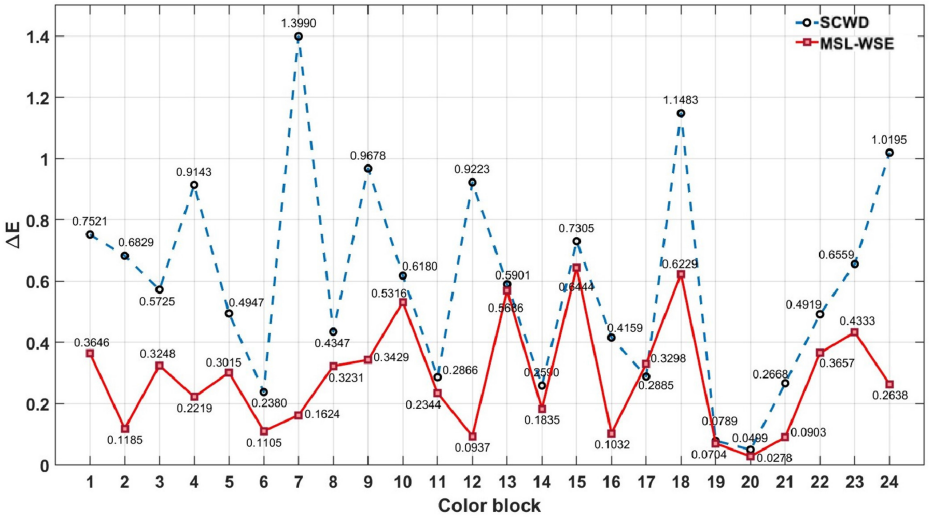

Fig. 11. Color difference evaluation results between retrieved colors and true colors obtained through the SCWD and MSL-WSE methods, where 1–24 represents the number of color blocks.

TABLE IV
EFFECTIVE EXTRACTION RATES OF SCWD AND MSL-WSE METHODS

Methods		Wavelength			
		475nm	539nm	640nm	820nm
SCWD	Sampling points	15900	18608	55518	57534
	Effective extraction rate	26.50%	31.01%	92.53%	95.89%
MSL-WSE	Sampling points	52070	57729	58275	58956
	Effective extraction rate	86.78%	95.38%	97.12%	98.26%

To further quantify the performance of weak echo detection, a detailed comparison list of the optimized waveform parameters between the SCWD and the MSL-WSE is presented in Table III. N_1 and N_2 represented the true and incorrect waveform identifications, respectively. The results showed that the SCWD method could only extract the waveform parameters from the high SNR channels. Among them, the central position at N_2 of the 820 nm channel did not correspond to the parameters at the N_2 in other channels, indicating that a false detection had occurred. In contrast, the MSL-WSE method extracted waveform parameters close to the true values in all channels, overcoming the limitations of insufficient information from a single channel. The MSL-WSE method demonstrated significant improvements compared to the SCWD method.

The above description primarily analyzed the results of weak echo detection from the waveform perspective. However, in the MSL system, the final extraction results were reflected in the true-color 3-D point cloud. Therefore, Fig. 9 further displayed the effects of the SCWD and MSL-WSE methods in extracting weak echo signals through the view of point cloud visualization. Comparatively, the presented method improved the accuracy of RGB values through waveform parameter optimization, resulting in a more uniform and realistic color representation of the point cloud [see Fig. 9(c)]. In addition, the MSL-WSE method effectively reduced noise points and enhanced the point cloud features of low-reflectivity objects. The boundaries of the "marmot" and the green beam were more complete, even the details of the millimeter level "拉萨" were displayed with clearer details (white box). The shape and structure of the picture frame were also more regular [red box in Fig. 9(f)]. This revealed that the MSL-WSE method obtained superior echo parameters and enhanced the data quality of the point cloud.

In Fig. 10, the accuracy of these two methods in weak signal extraction was quantitatively evaluated using linear evaluation metrics, color difference, and effective extraction rate. The results showed that the MSL-WSE method significantly improved the R^2 of each channel and reduced the RMSE. Particularly in the shortwave band (blue band), the R^2 increased by 10.7% and RMSE decreased by 52.63%. Besides, in Fig. 11, the MSL-WSE method effectively reduced the color difference between the point cloud and the real scene, making it closer to the true color of the color checker. Specifically, in the 24th and 7th color blocks with lower reflectance, the color difference generated by the MSL-WSE method was reduced by 74.13% and 88.40% compared with the SCWD method. This signifies the superior performance of the MSL-WSE method in accurately capturing and representing colors in low-reflectance regions.

Moreover, 60 000 pulses were used to calculate the effective extraction rate of weak echoes in the mural, as shown in Table IV. Compared to the SCWD method, the MSL-WSE method significantly improved the effective extraction rate in all channels. Particularly, in the shortwave 475 nm channel, the SCWD method had an effective extraction rate of only 26.50%, while the MSL-WSE method increased it to 86.78%, as presented in Fig. 9(f) with a more detailed texture in the point cloud. This further indicated that the MSL-WSE method has higher detection capabilities in the low-energy shortwave channels, which is crucial to enhancing the effective detection range of MSL.

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In conclusion, the MSL-WSE method showed better performance in weak echo detection and waveform parameter extraction, which improved the visualization effect of point clouds by optimizing waveform parameters, resulting in significant improvements in quantitative evaluation indicators. The results showed that the MSL-WSE method had important application value in extracting weak signals and accurately obtaining target information.

V. Conclusion

In this study, we proposed an MSL-WSE method and demonstrated its advantages in detecting and extracting weak echoes through visualization of point clouds in color squares and murals. Compared with the SCWD method, the MSL-WSE method adaptively selected thresholds and filters, taking into full consideration both the characteristics of weak echo signals and the correlation between multiple receiving channels, then established confidence channels and confidence intervals to accurately detect and extract waveform parameters. The results indicated that within the low-energy channel (475 nm), the R^2 between the extracted colors and the true colors increased by 10.7%, the RMSE decreased by 52.63%, and the effective extraction rate significantly improved by 227.55%. For the seventh color block, which exhibited low reflectance and tended to produce weak echoes, the extraction of waveform parameters presented some challenges. However, the proposed method still effectively reduced color discrepancies by 88.40%.

In conclusion, the proposed MSL-WSE method has made significant improvements in detecting and extracting the waveform parameters of weak echo signals. Specifically, different filters and fitting models can be selected based on the characteristics of weak echo signals to minimize noise interference and extract effective waveform parameters. Besides, the MSL-WSE method can also be applied to airborne hyperspectral LiDAR to improve the detection accuracy of remote targets in low-energy channels and reduce the color difference between the point cloud and the real scene.

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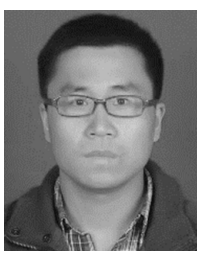

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